Expanded Topic E: measurement inputs

Five originals, the whole common theory, and five applications

This is a proposed input audit, not a compression result. The complete scientific edition and its bibliography successor have distinct scientific acceptance. The extraction and cost boundary still require independent input acceptance. No factor or additional source credit is asserted here.

The primary coordinate is the untrimmed PDF extraction, normalized to NFC, counting characters other than whitespace. It includes title matter, abstract, body, tables, references, arXiv banners and page furniture. Every physical page is included. The table reports that coordinate; it does not infer full graphical or computational effort from text length.

ID	Complete document	Pages	Characters
L079	PGL regular polytopes	9	21,212
L083	PSL regular polytopes	14	20,904
L021	PSL maniplexes	11	29,594
L056	Chiral polyhedra	8	15,871
L043	Chiral projective groups	25	52,705
U	Whole common theory	20	57,473
A1	L083 application	2	2,922
A2	L079 application	1	1,858
A3	L021 application	5	14,452
A4	L056 application	2	4,144
A5	L043 application	4	8,961
REQUIRED	Required programs and data	20	24,904
VALIDATION	Optional validation programs	7	13,974

There are 67 source pages, 34 scientific replacement pages and 27 program-appendix pages: 128 input pages altogether. The whole 20-page common paper is paid once. A1 belongs to L083 and A2 to L079; the reversed numerical order of those source identifiers is intentional. The two code appendices are distinct whole-file alternatives to the raw files, not additional copies of them.

The later written comparison must use all five source manuscripts and the whole common paper plus all five applications. A conditional source/application comparison is meaningful only after acquisition of the entire common theory, all six required finite-proof programs, the mandatory JSON and retained mathematical inputs. Other topics supply no free common theory.

Software, evidence and symmetry

Complete raw-file category	Files	Bytes
Six required finite-proof programs	6	27,832
Mandatory complete witness JSON	1	3,982
Six optional validation programs	6	15,153
Operational science/edition drivers	5	30,900
Scoped formal files	7	33,350

The six required programs establish finite assertions used by the written proof. They retain their required status. The 3,982-byte JSON is part of their evidence, and is printed in full with them in the 20-page required appendix. The six other programs are optional validation and source-correction illustrations; their whole seven-page appendix is separately disclosed. Each exact raw file and each complete printed block is checked without pruning diagnostic lines.

Optional validation is excluded on both sides in the written comparison. None of the five source PDFs supplies a complete executable listing. They do contain mathematical tables, finite presentation calculations and assertions using GAP or earlier catalogues; these remain in the manuscript counts. Unprovided historical software has unknown cost, not zero cost. A software-inclusive factor is therefore not supplied by this input packet. Full printed and raw alternatives are available as separate vectors for review; they must not be added together or treated as symmetric historical software costs.

The five operational drivers are recorded separately. Seven Lean files supply 42 scoped declarations (nine new and 33 reused). They do not formalize the full projective classifications, all finite computations, retained classical inputs or all source results. This measurement run neither recompiles Lean nor executes GAP. It reuses the accepted scientific scope and exact evidence hashes.

Exact source and result boundary

L079 is the nine-page arXiv manuscript carrying its 2009 banner. L083 is the 14-page file carrying both a 2006 arXiv banner and an August 2018 title-page date; no substitute edition is inferred. L021 is the 11-page author manuscript dated 15 May 2023. L056 is the complete eight-page authored PDF. L043 is the 25-page arXiv1606.08017v1 manuscript of 26 June 2016, despite the archival filename ending in “2017”; no journal collation is claimed.

The exact science map retains 54 earlier and 56 new dispositions. All five principal contracts are recovered, with literal corrections stated. L079’s additional 17-case semilinear census is outside its principal contract. L043’s quoted broad Theorems 8 and 10 and motivating census Tables 1–2 are not regenerated. Its four-subgroup correction to Lemma 5.12 and the missing residue class in the introductory formulation are preserved. L056’s auxiliary qualifications and the earlier L021 corrections remain explicit. These qualifications do not prune the complete source numerator. Prior question work and source authorship remain attributed; no new question credit follows from this audit.

Extraction, calibration and limits

All 30 coordinates are retained: three extraction variants, two normalizations (NFC and NFKC), and five units. The units are whitespace-delimited tokens, alphanumeric runs, nonwhitespace characters, characters after whitespace collapse, and UTF-8 bytes after collapse. Printed line-break hyphens remain; no semantic rewriting removes them. The selected variant removes only exact, bound running headers and isolated folios. L056 has no such removals. The calibrated selected variant adds only the declared occurrence-specific glyph changes.

The actual PDF dictionaries and effective encodings, not font names alone, identify every source event. The audit binds 92 source font dictionaries and 2,086 encoded glyph classes. Nineteen classes account for 279 calibrated occurrences: visible mathematical alphabets, eight matrix parentheses, five brace pieces and ten mapsto starting strokes. The parentheses replace control characters. Brace components are represented individually. The mapsto font glyph is only a leading vertical stroke with a short right stub: a declared vertical-piece proxy is used while the separately printed right arrow is left unchanged. These pieces are not joined into reconstructed formulas. All unaltered raw variants remain available.

Thirty-one exceptional outlines were viewed directly, including the twelve classes intentionally left unchanged. Six standard-font dictionaries are not embedded; their actual effective encodings are recorded without a claim of embedded-outline evidence. Every visible whitespace-decoded glyph is audited separately from generated spacing; none occurs in these source files. Ordinary ligature strings and correctly extracted phi shapes remain unchanged.

All 34 pages of L079, L083 and L021 were freshly viewed for this audit. The exact earlier complete visual readings of the unchanged L056/L043 source PDFs and all 61 replacement pages are explicitly reused. The bibliography successor changed three last pages; those pages were freshly viewed and separately reviewed, while other page texts remained identical. The visual record distinguishes these observations rather than calling every view fresh.

The PDFs contain inline images and vector rules, including diagrams and formula components. Their complete per-page inventory is retained. Extracted labels do not measure layout, strokes, incidence, image content or construction. Full graphics costs, outside classical proofs, original atlas construction, unavailable historical executables and complete source formalization remain null. No claim of full visible-text reconstruction or a complete proof-complexity scalar is made.

Replay. Run `check.py --study-root /path/to/study --out /fresh/path --render`. It re-extracts the exact supplied PDFs, reconciles actual font events, checks whole-file appendix inclusion, reproduces all 640 PDF/raw metric cells, exercises corruption controls and rebuilds this reader. Primary text, font programs and source rasters are not exported.